CJ Petka

ENGINEERING PORTFOLIO

Designed in CAD. Built by hand. Tested for real.

Mechanical Engineering and Computer Science
Northeastern University · December 2029

PROJECTS

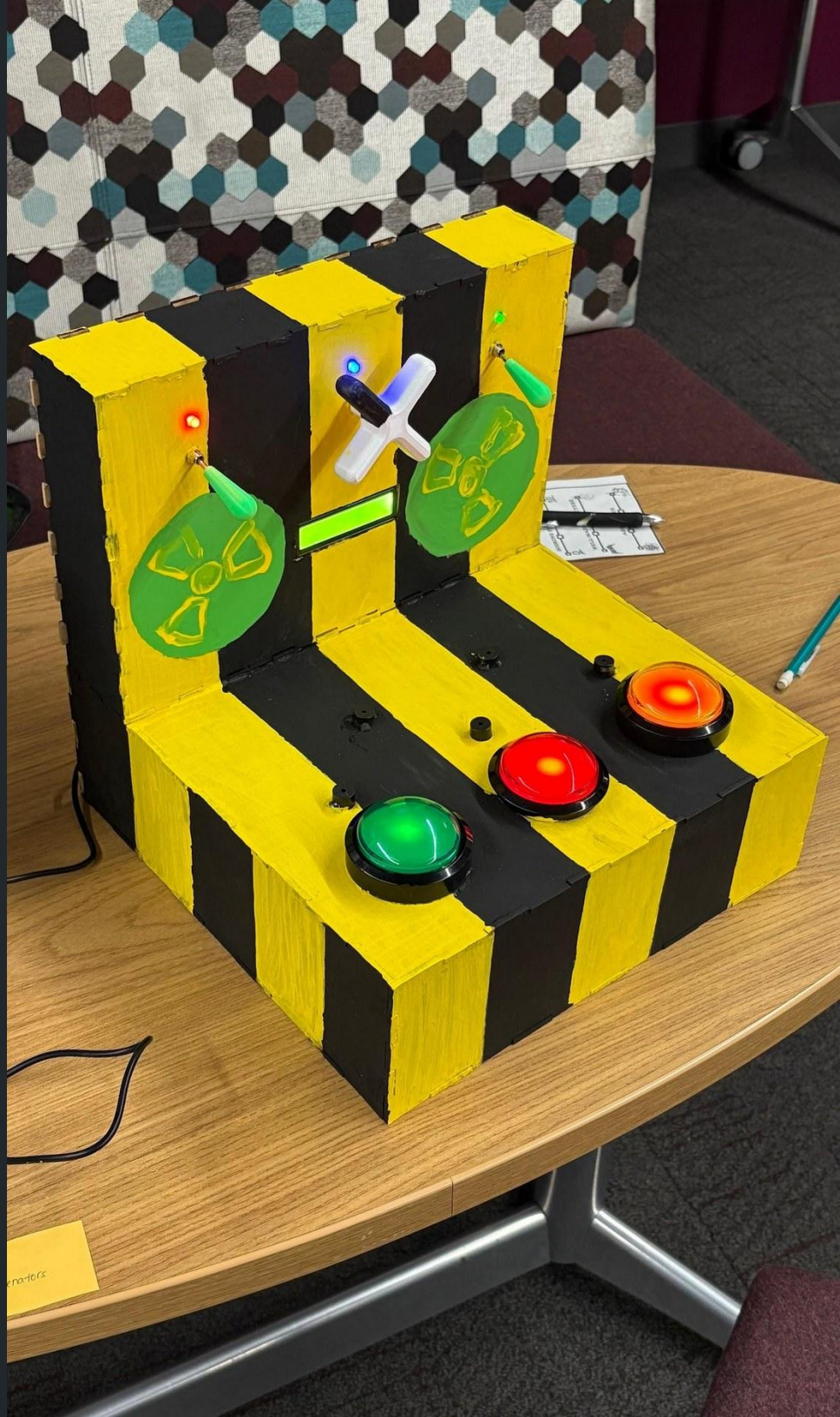
01 Steel Rail and Trolley Monitor Transport System

02 React to the Pressure

03 Drone Package Delivery System

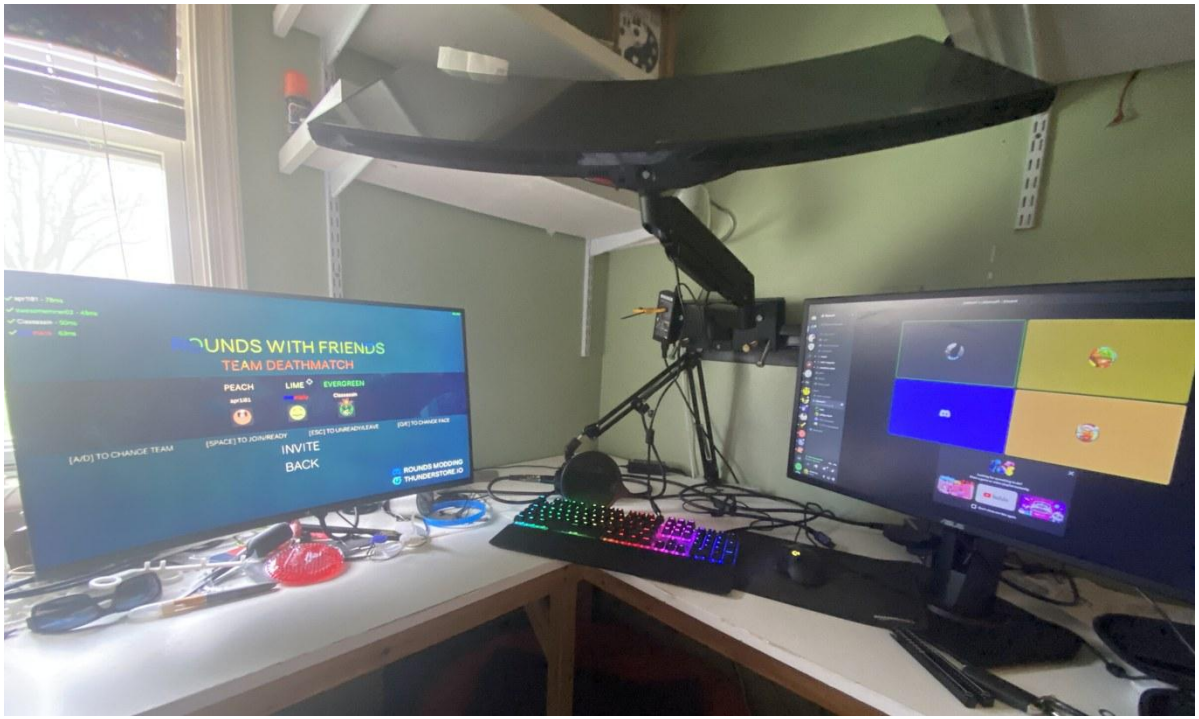
04 Spring-Powered Competition Vehicle

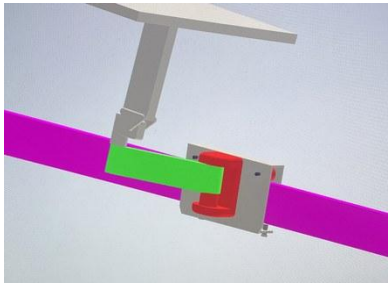
Boston, MA · (717) 569-4354 · petka.c@northeastern.edu · linkedin.com/in/cjpetka



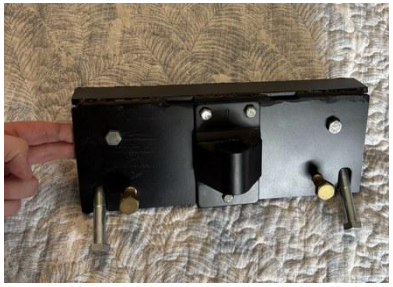
01 Steel Rail and Trolley Monitor Transport System

Self-developed Project · Spring 2024





Inventor model



Trolley face



Wheel in the rail

13 ft

steel span

100+

MIG welds

GOAL

Carry a VESA-mounted monitor along a fixed 13 ft span.

WHAT I DID

- Modeled the rail and adjustable trolley and produced technical drawings in Autodesk Inventor
- MIG welded the steel structure across more than 100 welds
- Turned custom trolley wheels on a lathe

RESULT

Installed and tested the finished system.

TOOLS

Autodesk Inventor · MIG welding · Lathe · Steel fabrication

02 React to the Pressure

GE 1501 Cornerstone of Engineering 1 · Team of 4 · Fall 2025

6
player inputs

$7.50
materials cost

GOAL

Build an engineering-failure carnival game for the Fall 2025 Expo. Ours is a Chernobyl-style control panel: shut off the blinking lights before the reactor melts down.

WHAT I DID

- Laser cut the enclosure and drilled mounting holes for the inputs
- Designed and 3D printed the valve and switch handles
- Soldered and repaired wiring, then debugged the Python game code on the Pico

RESULT

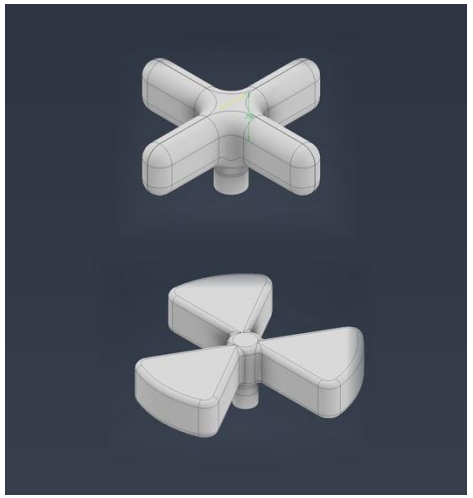
Demonstrated the finished game to the public at the Engineering Expo.

TOOLS

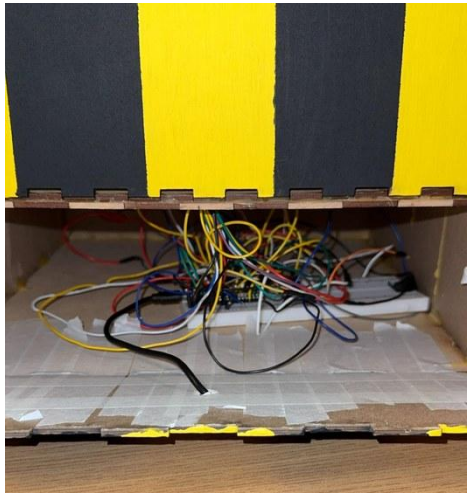
Laser cutting · 3D printing · CAD · Soldering · Raspberry Pi Pico · Python



Finished game



My handle designs (CAD)

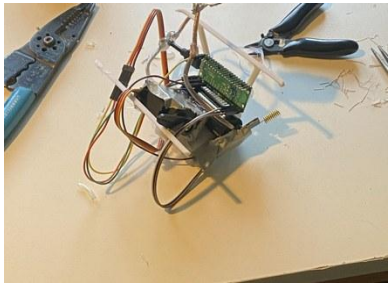


Wiring inside the enclosure

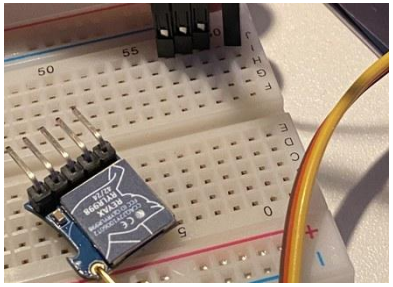
03 Drone Package Delivery System

Self-developed Project · Spring 2025





Electronics assembly



Radio module



DJI Mini 2

GOAL

Drop a small package from a DJI Mini 2, which has no built-in payload release, without relying on the drone's own controller.

WHAT I DID

- Designed a servo-actuated release mechanism and airframe mount sized to the drone's payload margin
- Programmed paired Raspberry Pi Pico microcontrollers to trigger the release over shortwave radio

RESULT

Remotely airdropped a small wheel of cheese as part of a promposal.

TOOLS

Raspberry Pi Pico (×2) · Servo actuation · Radio link · CAD

04 Spring-Powered Competition Vehicle

Engineering Competition · Fall 2023

76 m
school record

1:144
gear reduction

GOAL

Build a spring-powered vehicle that travels as far as possible.

WHAT I DID

- Modeled and 3D printed the vehicle with a 1:144 compound gear reduction
- Tuned printed gear tolerances so the gear train meshed reliably at high reduction

RESULT

Set the school distance record at 76 m.

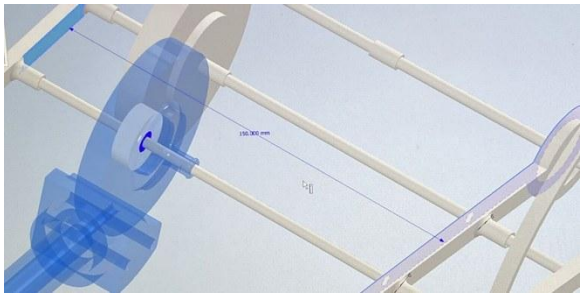
TOOLS

CAD modeling · 3D printing · Gear train design





Sliced drive gear



CAD model